

Pollinator module

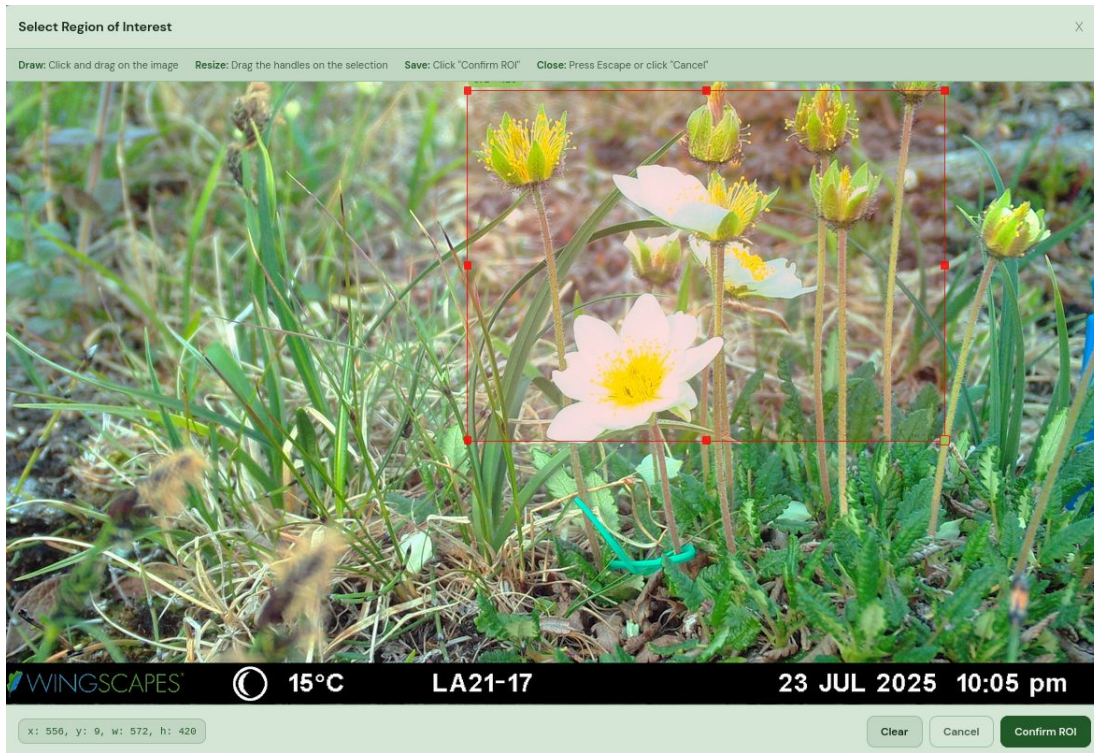
Running an inference run

Typing in a name for the run is mandatory (1). For detection **0.2** is standard, if you lower it to **0.05** the motion-based detection pipeline will detect a lot of more noise. Once models become better (higher recall results) this threshold can be raised.

You can decide to set a region of interest, ROI (2) that will open a pop up window. Clicking on “Start detection” (3) will take you to a popup-window shown on next page, if ROI box is checked.

The screenshot displays the 'Detect' step of the Pollinator module interface. At the top, a progress bar shows four steps: 1 Upload, 2 Detect (active), 3 Review, and 4 Export. The 'Run details' section contains a 'Name' field with the value 'hdd_2_gropen_dryo_pl_101_wsct', marked with a circled '1'. The 'Detection settings' section is divided into two columns. The left column, 'Full-Image Detection (YOLO)', includes a 'YOLO' dropdown set to 'yolo-v1 (active)' and a 'Confidence' slider set to '0.2'. The right column, 'Motion-Based Detection', includes 'Step 1: Insect Detection' set to 'binary-v1 (active)' and 'Step 2: Insect Classification' set to 'group-v1 (active)', both with 'Confidence' sliders at '0.2'. Below these, a checkbox 'Use manual ROI' is checked and marked with a circled '2', with a 'Start at image' dropdown set to '1'. An 'Advanced' link is visible. At the bottom, a file upload area shows '120 images selected · 161.8 MB' and a 'Start detection' button marked with a circled '3'. The interface also features 'Files' and 'Folder' tabs and a dashed box for image uploads with the text 'Drop images here or click to browse' and 'JPG, PNG -> Up to 12,000 images per run'.

This window will let you draw a rectangle, nothing outside of it will be caught.



The advanced section on the upload page:

Affect both pipelines:

- *Skip flash/foggy frames*: skips images if flash taken or it is too foggy
- *Large-motion tile overlap*: important when the two pipelines merges detections. Lower means fewer duplicated boxes that overlap each other. Tied to “Detect large motion” checkbox.

Only has effect on motion-based pipeline:

- *Crop padding*: adds a bit extra context around each bbox
- *Min/max contour area*: smallest and largest areas detected
- *Background sample size*: samples 0 (default) images to use as a median image that is used to compare first image against something. Turned off for now as it makes the inference run take longer in larger datasets.

▼ Hide advanced

Crop padding ⓘ	Min contour area (px²) ⓘ	Max contour area (px²) ⓘ
0.3	400	35000
Background sample size ⓘ	Large-motion tile overlap ⓘ	
0	0.35	
☒ Skip flash frames ⓘ	☒ Skip foggy frames ⓘ	☒ Detect large motion ⓘ

Defaults match the production pipeline. Only change if a specific batch needs tuning.

When images has been uploaded you are automatically taken to the **Detection page** where you find the progress. Today that process is around 100 images per 5 minutes, but it does vary depending on if using ROI and the confidence score used. It will say “completed” when it is done but you can go into the **Review page (4)** or **(5)** immediately and start reviewing if you want to - just know that there is no auto-refresh built in, so you will have to manually refresh that page to get updated images and crops shown.

✓ Upload

2 Detect

3 **5** Review

4 Export

COMPLETED

Created 6/3/2026, 8:12:17 AM

120 of 120 images processed (100%)

Completed in 6 min

4
Open review →

Detections so far · 564

By class

Fly

442

78%

Other

52

9%

Butterfly

44

8%

Bumblebee

26

5%

By source

Classical preprocessing

409

YOLO

107

Both

48

Recent activity

03:30 PM

Run complete: 564 detections across 120 images.

03:24 PM

Sampling background...

03:24 PM

Loading models...

03:24 PM

Starting run on 120 images (from image 1).

Configuration used

► frozen at run start

In the **Review** page there are several ways to select crops.

Selecting crops:

You can click on a box directly in the image (6) or drag to select several boxes (7). You can also select crops directly in the right column, either through normal click, shift-click or ctrl/cmd-click (8).

To the bottom right you can confirm what label the selected crop(s) should get (9) (10).

The screenshot shows the WingScapes Review interface. The main image is a photograph of a white flower with yellow centers. A dashed black box labeled "ROI section" is drawn around the flower. Two orange circles with numbers are overlaid on the image: circle 6 is on a small red bounding box around a fly, and circle 7 is on a blue dashed bounding box around the entire flower. The interface includes a top navigation bar with "Upload", "Detect", "Review" (active), and "Export" buttons. A left sidebar shows a list of image thumbnails. The right sidebar, titled "CROPS 4", lists four crops with their YOLO scores and labels: Fly (0.84), Fly (0.23), Fly (0.59), and Fly (0.54). The bottom panel shows "PREDICTIONS" for the selected crop, "THRESHOLDS" for Full-Image and Classification, and a "LABEL" dropdown menu with options: Fly, Bumblebee, Butterfly, Other, and Background (reject). The "Confirm" button is highlighted.

This close-up shows the "LABEL" dropdown menu. The "Fly" option is selected, indicated by a blue dot. An orange circle with the number 9 is over the dropdown arrow. Below the menu is an "Unsure" button and a "Confirm" button. An orange circle with the number 10 is over the "Confirm" button.

This close-up shows the "CROPS 4" list. The first crop, a fly with a YOLO score of 0.84, is highlighted with a red border. An orange circle with the number 8 is over the crop's selection icon.

Using keys to quickly label crops:

Left and **right arrow** will let you move through crops in an image, left to right.
Up and **down arrow** lets you go through the list of images.

You can type **1-4**, **x**, **d**, **y** and **u** for quick labeling and shortcuts.

1 = Fly

2 = Bumblebee

3 = Butterfly

4 = Other

x = Background

u = Unsure (will not label at all)

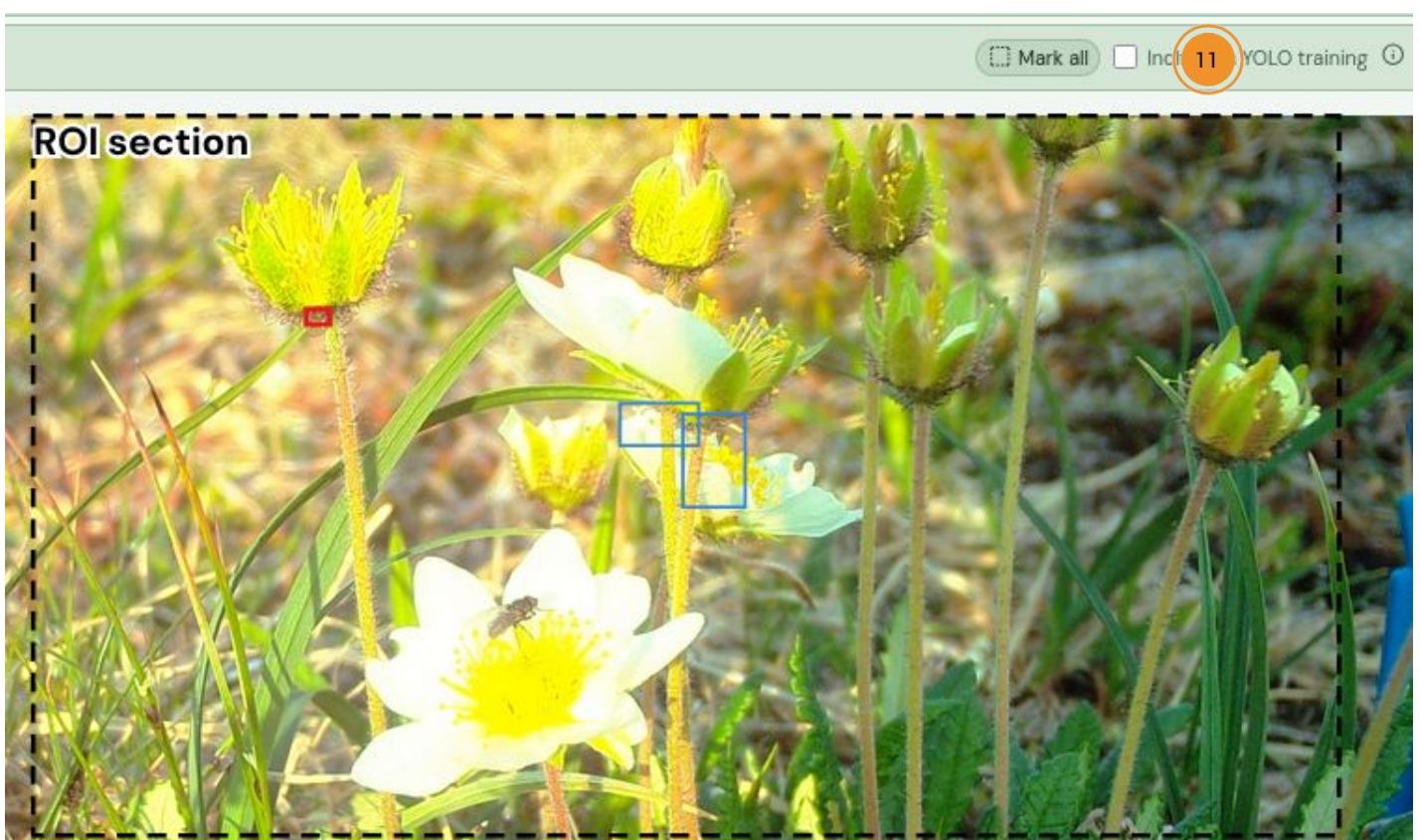
y = Include image in YOLO training

d = remove selected image from slide (all visible crops still in the image are labeled as Background)

You can batch label anything that is selected. In case you make a mistake there is a rewind option with **ctrl-z**.

Selecting crops for retraining of pollinator models

Crops that you label will automatically be available to retrain the motion-based models, but not automatically for YOLO. In order to get quality data for the YOLO model to learn on you specifically select the “Include in YOLO training” **(11)** checkbox. Make sure that all pollinators on those images are annotated, as that image is treated as ground-truth.

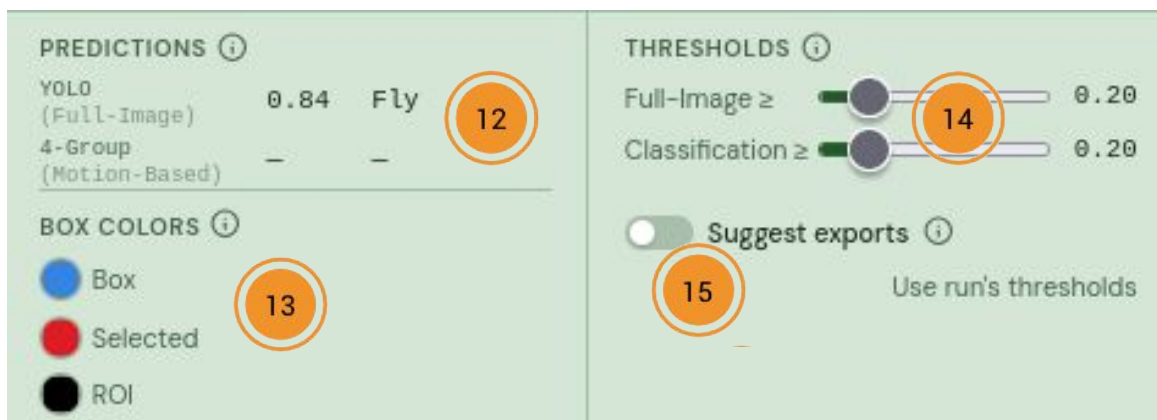


Reading the data and adjusting thresholds

In (12) you will see how much confidence each model has to what label it has set. Often the models complement each other. Where YOLO missed a pollinator the Group classifier may have found one. If they both agree with high certainty that it is a fly then that is likely also the case.

(13) decides what color all boxes should have on the **Review page**. You may need to adjust these colors depending on what plot is explored, as they vary in color.

You can tinker with thresholds in (14), this can be useful once the models has been trained with lots more data and reaches a recall rate near 0.80-0.90 at a threshold over 0.5. Then it also pairs well with (15) where everything over the thresholds limits are moved to the **Export page** (those crops are shown blue and you have an option to accept or remove them there).



You can go to the **Export page** by clicking the stepper (16).



Export page

In the **Export page**, what is green is confirmed by you as a user. When there is overlap, the larger box is marked red (17) and will not be exported unless you as a user click on it and mark it as OK on the right column, or directly on a box line in the image. This is in place so the tool does not count a pollinator more than once.

Images in the slider are **green** if all OK or **red** if there is at least one overlap. If you have enabled the “Suggest exports” toggle in **Review page**, all detections over the threshold limit are exported to this page and shown as **blue**.

“Merge overlap” is related to intersection over union (IoU), lower value here will merge more aggressively (outer rectangle will be red). Also worth noting that if one rectangle has its middle point inside another rectangle, that rectangle becomes red (18).



Exporting CSV gives two options, either “per image” which will include counts of all insect types, or “per crop”, which will include confidence scores per crop.

Exporting crops will download a folder with all crops separated into subfolders based on what type they belong to, which is useful if you wish to train models outside of the application with Jupyter Notebooks.

Annotations can also be exported, it will save images with drawn boxes and ROI if it exists.

Review layout has two different views: “image-first” and “crop-first”. Image-first is the newest edition and in general what you want to use. Using crop-first will get really messy if you are running on 1000+ images, but it can be useful to filter detections into classes.

You can also change key shortcuts if you want to set it up in a way you prefer.

Review layout

Pick how the review page is organized. Both layouts show the same data, the difference is whether you step crop-by-crop or image-by-image.

Image-first **ACTIVE**

Scrollable rail of source images on the left. Selecting an image shows it large with all its crops. Best when context inside the image matters.

Crop-first

Grid of crops grouped by review status and class. Selecting a crop shows its source image with sibling bboxes faded out. Best for tight, high-throughput review.

Reset layout to default

Review shortcuts

Single-key shortcuts for the review page. Click a key, then press the key you want.

Classify: fly	1
Classify: bumblebee	2
Classify: butterfly	3
Classify: other	4
Reject (background)	X
Mark unsure	U
Include image in YOLO training	Y
Reject image's unreviewed crops	D

Reset shortcuts to defaults